

25AM202 DESIGN THINKING AND ENGINEERING ORIENTATION

Hours Per Week :

L	T	P	SL	C
0	0	2	0	1

PREREQUISITE KNOWLEDGE: Nil.**COURSE DESCRIPTION AND OBJECTIVES:**

This course introduces students to the fundamentals of design thinking and its applications in engineering problem-solving. It emphasizes empathy, creativity, and iterative prototyping as essential skills for mechanical engineers. The objective is to orient students toward human-centered design, sustainability, and innovation in engineering practice.

MODULE-1**UNIT-1****0L+0T+12P+0SL = 12 Hours****INTRODUCTION TO DESIGN THINKING:**

Principles and stages: Empathize, Define, Ideate, Prototype, Test. Divergent vs. convergent thinking. Case studies: Engineering innovations inspired by design thinking.

UNIT-2**EMPATHY & PROBLEM DEFINITION:**

Tools: Empathy maps, journey maps, stakeholder analysis. Defining problem statements using “How Might We...?” questions. Identifying user needs in campus and workshop contexts. **Karuna** and **Sahridaya**.

PRACTICES:

- Conduct an empathy exercise with peers to identify problems in classroom ergonomics.
- Prepare a problem statement based on observations from campus transport
- Prepare a problem statement based on observations from campus workshop/lab.
- Redesigning Classroom Furniture for Ergonomics – Empathy with students & faculty.
- User-Centered Design of Bicycle Components – Comfort, efficiency, and sustainability.
- Defining Problems in Campus Transportation – Identifying pain points in buses/shuttles.

MODULE-2**UNIT-1****0L+0T+18P+0SL = 18 Hours****IDEATION & BRAINSTORMING:**

Ideation methods: SCAMPER, brainstorming, morphological charts. Generating multiple alternatives for mechanical engineering problems. **Kalpnaand Pratibha**.

UNIT-2**PROTOTYPING FOR MECHANICAL ENGINEERS:**

Low-fidelity prototypes (paper, cardboard, clay models). High-fidelity prototypes (basic CAD, 3D printing, simple workshop fabrication). Prototyping examples: Bicycle component redesign, tool modification, lab equipment enhancement. **Prayogaand Yukti**.

UNIT-3**TESTING & ITERATION:**

Methods for user testing and feedback collection. Iteration cycles for improvement.